



OPEN Effective skin cancer classification by modified and optimized inception-ResNet-V2 model

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Artificial Intelligence tools are flourishing in biomedical diagnosis, particularly in oncology. The prediction of skin cancer from dermoscopic images using deep learning neural networks has gained importance in recent years because of their inherent early non-invasive diagnostic capabilities. This study presents the results of the classification of benign nevus and malignant melanoma lesions using a deep learning model. The model incorporates an efficient pre-processing stage powered by median filtering and a class-balancing stage powered by the Synthetic Minority Oversampling Technique (SMOTE) to improve classification results. First, the classification efficiency of four pre-trained models, namely, ResNet50, EfficientNet B0, Inception-V3, and Inception-ResNet-V2, were studied, and the results revealed that they achieved accuracies of 93.90%, 94.37%, 94.87%, and 95.77%, respectively. Second, the effect of optimization and hyperparameter tuning on the Inception-ResNet-V2 model is studied considering Adam, Nadam, and AdaMax optimizers, with fivefold cross validation. The experimental results revealed that the AdaMax optimizer with validation achieved an overall consistent performance with accuracy, sensitivity, and specificity of 97.65%, 96.67%, and 98.92%, respectively. The results support the efficacy of the model in predicting skin cancer malignancy; thus, its integration into clinical practice could benefit healthcare services.

Keywords Convolutional neural network, Inception model, ISIC 2019, Pre-trained models, Residual networks, Skin cancer, Process innovation

Skin cancer is characterized by the abnormal growth of skin cells. According to reports by the World Health Organization, the primary cause of skin cancer is exposure to ultraviolet radiation. Surveys depict that skin cancer is the 23th leading cause of death worldwide with highest mortality incidences registered in Europe¹. It is estimated that, 112,690 new skin cancer cases with 14,110 deaths in the US in the year 2025². These statistics reveal the impact skin cancer has and its geographical variations across the globe. Skin cancer is broadly classified into three types depending on the type of skin cells: melanoma (MEL), basal cell carcinoma (BCC), and squamous cell carcinoma (SCC). More than 100 subcategories of skin lesions have been identified to date. Depending on the severity of the disease, skin lesions are classified as benign or malignant. As with any type of cancer, skin cancer can be a life-threatening condition when it enters a malignant state. Therefore, early diagnosis of skin cancer can save lives.

The conventional diagnosis of skin cancer starts with visual inspection of the lesion by a physician, followed by biopsy if required. Biopsied tumours carry a wound capable of pro-metastatic progression which causes mortality if surgery is delayed³. Computer-aided diagnosis of skin cancer using dermoscopic imaging is a painless alternative. Therefore, it could be used as an early diagnostic tool. The conventional diagnosis of skin

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lesions based on the asymmetry, border, color, diameter, and evolution (ABCDE) rule for staging skin cancer has been adapted by many computer-aided design (CAD) tools for the effective classification of cancer stages^{4,5}. However, with the advent of artificial intelligence (AI) techniques, machines can learn the image features themselves, making diagnosis better. AI-powered automatic diagnosis of skin cancer could facilitate expert level diagnosis feasible in a non-invasive manner which is highly beneficial in clinical decision making⁶. The two main categories of AI algorithms are machine learning (ML) and deep learning (DL). ML frameworks demand a feature extractor block after preprocessing⁷. While a DL method can extract features efficiently when trained with sufficient data and thus are adequately investigated for implementing dermoscopic image based skin cancer diagnosis⁸. In specific, pre-trained models and ensemble learning based models could achieve the desired accuracy even with minimal data^{9,10}.

Significant challenges associated with the use of AI models for medical diagnosis relies in the insufficient, imbalanced training data, which often hurdles model performance as well as generalisation ability¹¹. Most existing methods concentrate on using augmentation to balance the dataset and leave diversification of dataset unattended. This limits model predictions when images acquired under practical setups—which may include artifacts and are often distorted. Such a limitation hurdles practical integration of model with smart device for enabling remote diagnostic facilities. The proposed method employs artifact removal and data augmentation that addresses class imbalance and ensures robustness under practical image acquisition. Thus, enables the model to generalise well with images acquired under practical conditions (when new images are distorted due to lighting or environmental conditions, acquired in different geometries). A robust modified Inception-ResNet-V2 model is chosen for classification which ensures robustness of predictions to unseen data. Further, model generalisation and robustness to unseen data are validated by fivefold cross-validation and hyperparameter tuning. Such a model generalisation and convergence become important when the proposed method is integrated with smart device based diagnosis systems. The objectives of the study are,

- To conduct a literature review of state-of-the-art in the classification of skin cancer from dermoscopic images using DL techniques.
- To develop a transfer-learning-based optimized Inception-ResNet-V2 to classify skin lesions into two classes. Initially, the dermoscopic images are pre-processed for artifact removal. The training dataset is augmented to address class imbalance and sample diversification. Modified Inception-ResNet-V2 model is defined and its model parameters are tuned by three different optimisers, fivefold cross validation and the best-performing model was chosen to classify the hold-out test data.
- The performance of the developed model is investigated by calculating its accuracy, specificity, sensitivity, and Area Under the Receiver Operating Characteristics (AUC-ROC).
- The performance of the developed model is compared with that of existing state-of-the-art DL methods.

The paper starts with an introduction to skin cancer and the role of AI in skin cancer diagnosis in Section “[Introduction](#)”, followed by a review of recent advancements in the AI-powered diagnosis of skin cancer in Section “[Related work](#)”. Section “[Materials and methods](#)” details the input dataset, model development, and testing methods. The workflow of the proposed methodology and architecture of the modified Inception-ResNet-V2 model used in the experiment are explained in Section “[Proposed methodology](#)”. The experimental results are presented in Section “[Results](#)”, and conclusions are provided in Section “[Conclusion](#)”.

Related work

The performance of an AI tool directly depends on the quality and quantity of the input data used for training. Thus, various pre-processing techniques adapted to improve dermoscopic image quality and their impact on segmentation are studied in literatures¹². Further, segmentation of skin lesions for improving prediction efficacy are also investigated. For instance, Res-UNet model were employed for skin lesion segmentation and demonstrated to achieve a Jaccard index of 77.2% while segmenting skin lesions¹³. In another experiment, grab-cut segmentation, Gray-level Co-Occurrence Matrix (GLCM)-based feature extraction and ML based classification was studied. And the experimental results demonstrated the efficacy of Support Vector Machine (SVM) predictions¹⁴. Further, a recent researches work employed Adaptive snake and region growing method for lesion segmentation and SVM for classification. Such a method was demonstrated to achieve improved accuracy of predictions¹⁵. SVM classifiers demonstrate improved model performance compared to other ML models¹⁶. Though machine learning methods could achieve the desired performance in classifying skin lesions, DL methods excel in making predictions with real-world data and providing reliable results^{17–20}. Specifically, convolutional neural network (CNN) models were demonstrated to achieve the desired prediction performance in case of image based diagnosis²¹.

Specifically, transfer learning techniques have emerged as a boon in the classification of skin lesions. For instance, DL models developed by adapting transfer learning from MobileNet-V2 combined with an LSTM network could classify the HAM-10000 dataset into three classes with an accuracy of 88%²². Numerous models have been developed by adapting transfer learning techniques, and have been found to be effective in classifying skin lesions²³. For example, the EfficientNet architecture showed an AUC of 0.968 in identifying melanoma and non-melanoma from the ISIC-2019 and ISIC-2020 datasets²⁴. While, AlexNet achieved classification accuracies of 96%, 97%, and 95% with the MED-NOD, Derm, and ISIC datasets, respectively²⁵. In addition, ensemble of pre-trained networks was found to improve classification efficiency. For example, the ensemble average of forecasts made by ResNet-152, Se-ResNeXt-101, NASNet, and DenseNet-161 was demonstrated to achieve, 80% accuracy and 98% AUC in classifying the DermNet dataset and 93% accuracy and 98% AUC in classifying the ISIC dataset²⁶. While, ensemble of ResNet-50 models achieved a top-1 accuracy of 93% in classifying the

HAM-10000 dataset²⁷. Experimental results show that modifying and tuning of pre-trained models can achieved excellent classification performance in comparison with readily using pre-trained models²⁸.

Despite the massive use of transfer learning techniques for skin cancer diagnosis tasks, custom deep convolutional neural network (DCNN) models have been developed and demonstrated to classify benign and malignant melanomas from ISIC datasets²⁹. For instance, a DCNN with dynamic kernels was shown to efficiently classify the HAM-10000 dataset into seven classes³⁰. A hybrid DCNN model with three unique tailored feature extractor modules was developed and demonstrated to classify three variants of the ISIC dataset, namely, ISIC 2016, 2017, and 2018, with a good AUC³¹. Potential directions of research pertaining to skin cancer diagnosis is primarily motivated towards development of combining patient meta-data with screening images in order to advance AI tools, from diagnosis to prognosis³². For instance, a Convolutional Neural Network—Artificial Neural Network (CNN-ANN) model that can identify malignant and benign melanoma from the shortlisted ISIC 2019 dataset and the corresponding patient metadata was developed³³. In addition, predictions considering multiple criteria were demonstrated to achieve excellent classification accuracies³⁴. Similarly, introducing Explainable AI (XAI) techniques with model are vastly investigated to enable AI acceptance in medical applications³⁵. For instance, Machine learning models add XAI in the form of permutation importance plots, partial dependence plots (PDP), and SHapley Additive exPlanation (SHAP) plots for skin cancer classification³⁶. In summary, although each AI technique has unique advantages in classifying skin cancer images, pretrained models and ensemble learning techniques have shown promising results. Additionally, future AI models for medical diagnosis will be capable of providing prognostic solutions based on patient metadata and justifying their decisions with the power of XAI.

Materials and methods

Dataset

The ISIC 2019 dataset is used in this study. The native ISIC dataset was downloaded from “<https://challenge.isi-c-archive.com/data/#2019>”. This dataset has eight different classes of skin lesions. From this dataset a subset of 1065 images were obtained with 600 images pertaining to malignant melanoma class and 465 images pertaining to benign nevi class. Along with these images a binary coded ground truth. “csv” file is used for training the model. This dataset is split in 80:20 ratio for training and testing the model. Stratified sampling is adapted for ensuring equal test and train split. Further, the training data is augmented for diversification of the input data that represent images obtained under various environmental conditions. The testing data is not augmented in order mimic real practical images.

Noise removal

Dermoscopic images are often corrupted by hairlines, air bubbles, ink markings, and black frames, which hinder effective image analysis. Thus, artifact removal techniques such as edge detection, inpainting, region expansion, and intensity thresholding are often used to improve image quality and acquire clearer images that better suit precise skin lesion investigation and identification.

Data augmentation

Medical image datasets are often imbalanced because of the rarity of the disease and availability of variants. Imbalanced datasets, when used directly for training AI models, result in biased classification towards the majority class, and thus seek attention. Class imbalances are addressed by different methods, such as (i) weighted class learning, where the loss function is weighted to give importance to the minority class, (ii) oversampling the minority classes, and (iii) under sampling or removing samples from the majority classes. However, removing the samples may lead to underfitting; thus, oversampling is typically adopted. Simple data duplication may result in overfitting; thus, data-interpolation-based synthetic data generation was adapted for oversampling minority classes.

Convolutional neural network

Convolutional neural networks work analogous to biological visual perception systems, where the visual cortex has cells that respond to smaller visual field regions called receptive fields^{37,38}. Each Convolution layer performs two operations: convolution of input (O_{i-1}) with kernel (W_i) followed by bias b_i addition and application of an activation function over W_i to obtain the output (O_i), which acts as input for the next convolution layer. A convolution operation is performed by applying a smaller kernel window and convolution of the kernel over the input. The convolution operation is given by Eq. (1).

$$\text{Convolution, } h(x) * g(x) = \sum_{u=-\infty}^{\infty} h(u) \cdot g(x-u) \quad (1)$$

where $h(x)$ is the kernel, $g(x)$ is the convolved input, and $*$ is the convolution operation. The intermediate output (Z_i) generated by the convolution layer is given by Eq. (2), whereas the output (O_i) generated after the application of the activation function is given by Eq. (3).

$$\text{Intermediate output, } Z_i = (W_i * O_{i-1}) + b_i \quad (2)$$

$$\text{Output, } O_i = \text{Activ}(Z_i) \quad (3)$$

In CNNs, neurons belonging to consecutive layers are not fully connected. Some neurons share weights, thereby simplifying learning and network convergence. Each convolution layer uses 'n' independent kernels to produce 'n' feature maps, which can be represented by Eq. (4).

$$\text{Output}, O_i^n = \text{Activ}((W_i^n * O_{i-1}^n) + b_i^n) \quad (4)$$

The pooling layer (average or max) down samples an image to capture the most important features and discard the least important ones. A fully connected layer contains neurons that are connected to all the activations in the previous layer. This layer performs the classification task using flattened data sent by the previous layer. The basic blocks that constitute a convolutional neural network is illustrated in Fig. 1.

Residual networks

As the neural networks deepened, the network's learning is affected by the vanishing gradient (VG) problem. Repeated multiplication of derivatives of activation function results in exponentially small gradients at the end of forward pass. This extremely smaller gradient when multiplied with the calculated loss results in a very smaller value which when backpropagated makes only minimal changes (than required) to model weights that directly slowed down or stopped learning process. However, skipping intermediate layers using identity blocks enables the network to learn the residual function instead of the direct relationship between the input and output. This arrangement helps overcome the vanishing gradient problem with deeper networks and aids in faster model convergence. Let the input be 'x' and the corresponding output distribution be h(x). The difference between h(x) and 'x' gives the residue Res(x). The Eq. (5) and (6) represent the relationship between residue Res(x), the input 'x' and output h(x).

$$\text{Res}(x) = h(x) - x \quad (5)$$

$$h(x) = x + \text{Res}(x) \quad (6)$$

Equations (7) and (8), along with Fig. 2, clearly show that a residual block learns the residue Res(x) instead of the actual output h(x) during network training, layers in a residual block, are either trained or skipped by a skip connection based on the backpropagated error. Thus, residual networks are capable of producing the best model for a given dataset. Schematic of a residual block with skip connection is shown in Fig. 2.

The input–output relationship of residual networks can be represented by Eq. (9), where the output of the nth residual block is the input to the (n + 1)th residual block:

$$x_{n+1} = H(x_n) + x_n \quad (7)$$

Identity Block: The input and output dimensions were identical for each identity block. It has three convolution layers, followed by batch normalization and rectified linear unit activation. The input of the block was added to the output of the third layer of the identity block. These are the basic building blocks of residual networks.

Inception networks

Inception networks, in simpler terms, can be defined as network of networks. An inception network is built using multiple inception modules. Each inception module learns the input features by applying multiple convolutional filters in parallel and concatenating their output. An important building block in the inception module is the

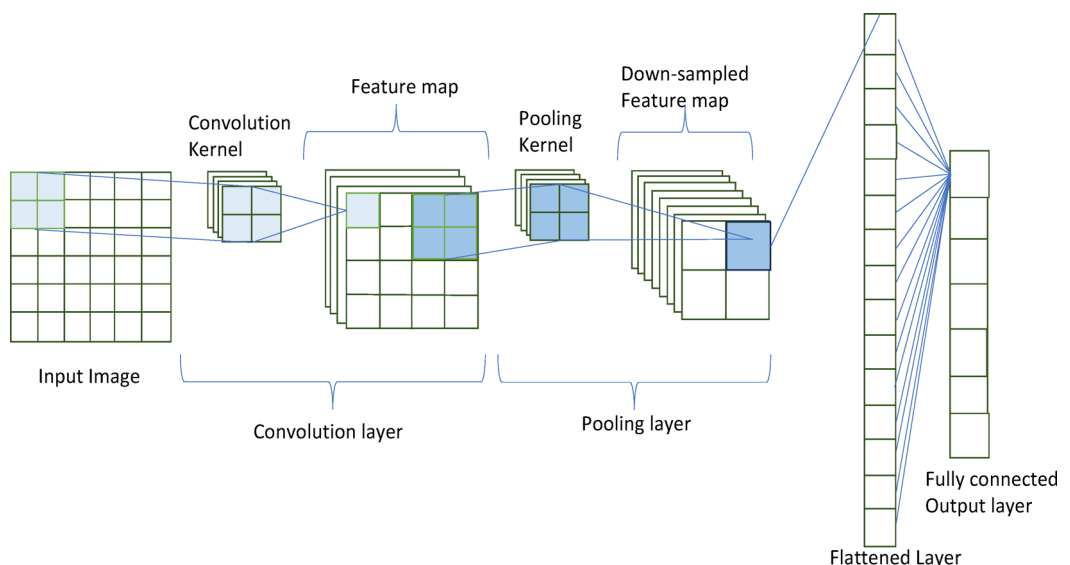


Fig. 1. Convolutional neural network architecture.

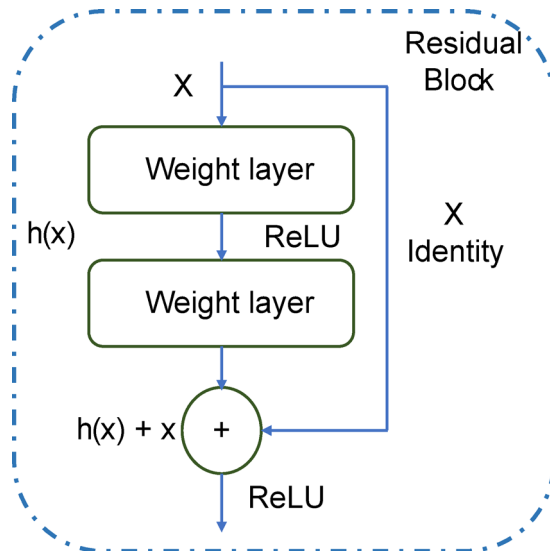


Fig. 2. Residual block.

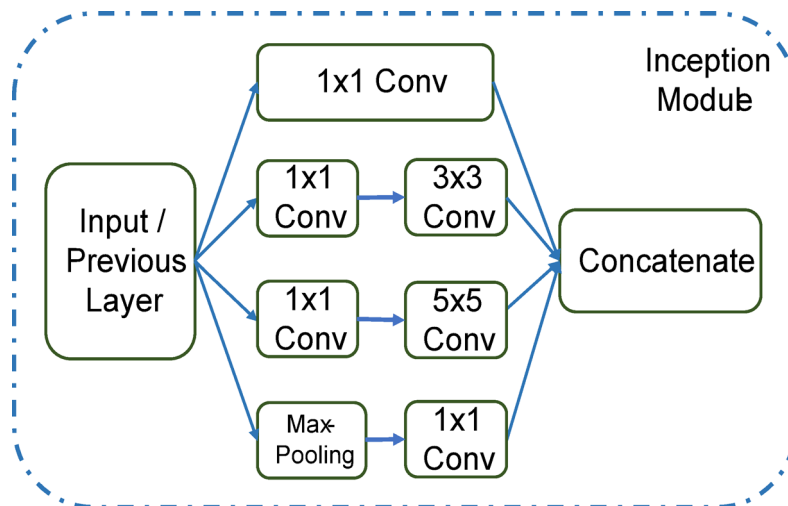


Fig. 3. Basic inception module.

1×1 convolution block, also known as the projection block, which performs channel-wise convolution, and thus, reduces the depth of the feature map. In addition, the module includes a max-pooling layer that learns the maximum value from a set of nonoverlapping regions of the input. Inception modules and, thus, inception networks simultaneously improve feature learning and model convergence, and avoid overfitting. The sample inception module is illustrated in Fig. 3.

Projection block: The input and output dimensions of the projection block differ. It has a convolution layer with a (1×1) kernel and stride (2×2) to change the input depth and match it with the output depth. This reduced the number of feature maps.

Model optimisation and hyperparameter tuning

Optimizing the model parameters using efficient optimizers and tuning the hyperparameters can result in a model with efficient classification performance. Three different optimization techniques based on stochastic gradient-based optimization, Nesterov accelerated gradient, and infinity norm-based optimization with fivefold cross validation were studied.

The most suitable set of parameters (weights and biases) for a neural network can be obtained by adapting a suitable optimization algorithm while training the network. In addition, the optimizers facilitate faster model convergence. Gradient descent algorithms are basic algorithms that update model parameters based on the negative gradient of the objective function. Stochastic gradient-based optimization algorithms are the preferred choice for many applications because their objective functions are composed of numerous subfunctions and are highly stochastic. The Adam optimizer is a stochastic gradient descent algorithm variant that uses first- and

second-order gradient moments to update model parameters. Weight and bias updates are performed using Eq. (10):

$$\phi_i = \phi_{i-1} - \alpha_i * \frac{m_{i(I)}}{\sqrt{m_{i(II)} + \varepsilon}} \quad (8)$$

where $m_{i(I)}$ and $m_{i(II)}$ are the first order and second order moment estimates, α_i is the step size and ϕ is the parameter being tuned. Adam optimizers merely follow a decreasing gradient irrespective of whether the next step is sloping down or ascending. This prediction (a lookahead property) of the next step is an advantage of the Nadam optimizer, which updates the weights based on the Nesterov accelerated gradient (NAG), which calculates the gradient of a predicted future parameter value against the current parameter value. It combines Adam with Nesterov moment to update the weights. The update function of the Nadam optimizer is given by Eq. (11).

$$\phi_i = \phi_{i-1} - \alpha_i * \frac{1}{\sqrt{m_{i(II)} + \varepsilon}} \left(\beta_1 m_{i(I)} + \frac{(1 - \beta_1) g_t}{1 - \beta_1^t} \right) \quad (9)$$

where β represents the exponential decay rate and g_t represents the gradient of the objective function. Adam's weight update rule scales the gradient in inverse proportion to the L^2 norm of the present and past gradients. The generalization of this rule results in an L^p Norm. When $p \rightarrow \infty$, this results in a simple and stable algorithm (AdaMax). The update function of the AdaMax optimizer is given by Eq. (12):

$$\phi_i = \phi_{i-1} - \alpha_i * \frac{m_{i(I)}}{(1 - \beta_1^t) * u_t} \quad (10)$$

where u_t denotes the exponentially weighted infinite norm. In addition, AdaMax has a simpler parameter update magnitude bound than Adam does.

Proposed methodology

Deep-learning-based architectures are efficient feature learners compared to ML-based models and often require minimal image preprocessing. The blocks that constitute the proposed Deep CNN based skin cancer classification model is illustrated in Fig. 4. This high level block representation enables interpretability to the black box model. The first step in the proposed DL framework is image preprocessing, which includes noise removal. After noise removal, the training data is balanced by data augmentation in order to avoid bias due to data imbalance in input classes. After balancing the dataset, input diversification was performed by flipping, shifting, and rotating the training images. This step enables adaptability of the model to practical environments. The Modified Inception-ResNet-V2 model was defined by customizing the output layers and retraining them with skin-lesion images. The specific significance of choosing this hybrid model is discussed in section “[Modified inception-ResNet-V2 architecture](#)”. The effects of optimization and hyperparameter tuning were studied using three optimizers, Adam, Nadam, and AdaMax, with fivefold cross validation. This further ensures model robustness to unseen data. After validation, the best-performing model is saved for making final predictions.

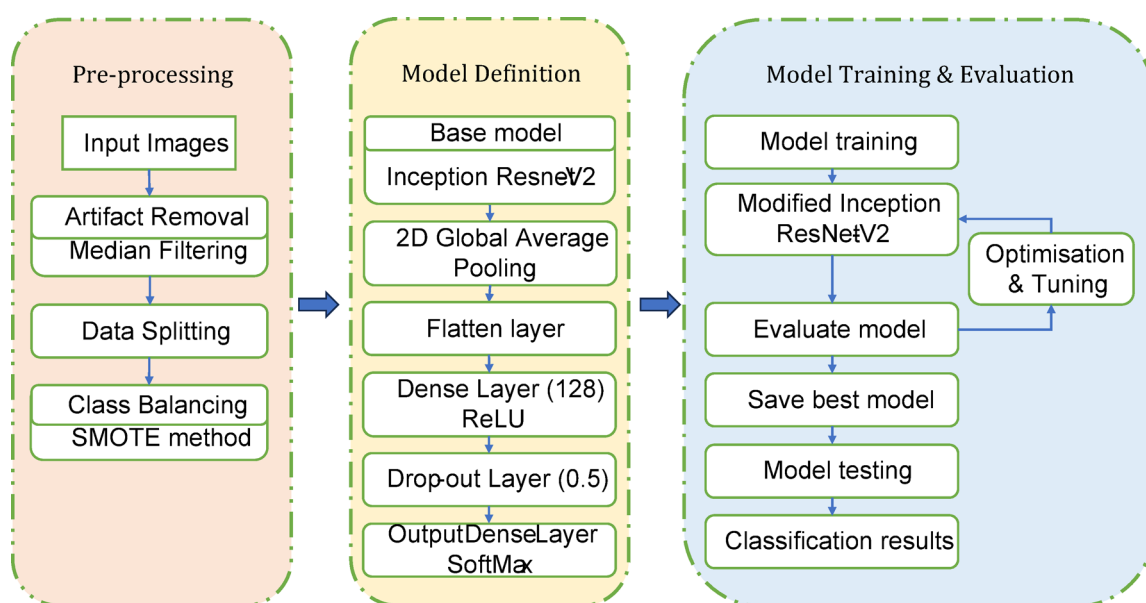


Fig. 4. Architecture/Block diagram of the proposed model.

Pre-processing & Data augmentation

One of the major sources of noise in dermoscopic images is hair lines and clinical markings, which hinder the efficiency of classification. These noises often appear as thin lines in dermoscopic images.

Artifact removal: Median filtering is employed for removing artifacts from the acquired images. Median filters, which are efficient in preserving edges while removing noise, can easily differentiate these dark line artifacts from the lesion boundaries. The median filter replaces each pixel with the median of that pixel and its neighbouring pixels. Thus, when a median filter encounters pixels belonging to the hairline or clinical marking (which are very thin), it replaces them with the median of their neighbourhood, resulting in the removal of dark lines by skin color. This improves image quality and facilitates efficient feature extraction, optimized model performance. However, for optimal results, the size of the neighbourhood window used for the median calculation (S_w) should be proportional to the image size. This was calculated using Eq. (13) for the optimal results.

$$S_w = floor \left(5 * \sqrt{\left(\frac{x}{768} \right) * \left(\frac{y}{512} \right)} \right) \tag{11}$$

where x and y are the input image sizes and constant 5 is the preferred window size for a 768 × 512 image. The images were resized to 299 × 299, as this is an optimal known choice for the Inception ResNet-V2 model. Artifact removed images are split in 80:20 ratio for training and testing. Stratified sampling is used as the input dataset is imbalanced. The training data set alone is augmented for class balancing and diversification while the testing set is kept as such for ensuring a more robust model evaluation. SMOTE technique is employed for oversampling the minority classes in the training set which resulted in a more balanced training data with 480 samples in each class. This ensures an unbiased training. The details on number of samples in each class are given in Table 1 below. This balanced data is further augmented to have 1000 samples in each class, by employing geometric transformations like shear, zoom, horizontal flip, rotation by 30°, and shifting. This augmentation for diversification is carried out to ensure the model generalises well with real-time data which may be affected by lighting, resolution of imaging device, or environmental conditions. This diversification carries significance while integrating model to practical clinical use. Further, it helps their integration to smartphone based diagnostic devices which is beneficial in resource constrained setting.

Modified inception-ResNet-V2 architecture

Inception-ResNet-V2 is a pretrained CNN model developed using inception blocks and residual connections. Residual blocks enable training deep neural networks efficiently by overcoming the vanishing gradients. Specifically, skip connections enable learning the residue that is needed to produce the required output. Further, these networks facilitate faster model convergence along with the desired accuracy. In addition, Inception blocks with parallel layers enable faster feature extraction by processing multi-scale features simultaneously. This facilitates extraction of both global and local features with reduced computational complexity. Residual connections ensure faster model convergence and avoid the vanishing gradient problem, whereas inception blocks avoid model ensure a balance between feature depth and efficiency by imparting parallel learning layers. Both, residual networks and Inception networks aim to achieve deeper learning—for accurate feature extraction, without overfitting, even with minimal dataset. Thus, the hybrid Inception-ResNet-V2, provide a powerful learning network, which is robust and generalises well to unseen real world data. The architecture of Inception-ResNet-V2 is shown in Fig. 5.

Inception-ResNet-V2 was built from three variants of the Inception-ResNet blocks: Inception-ResNet-A, Inception-ResNet-B, and Inception-ResNet-C. The basic blocks that constitute each of this variant is presented by separate blocks in Fig. 5, where Inception-ResNet-A is represented by mild yellow, Inception-ResNet-B is represented by mild green, and Inception-ResNet-C is represented by mild blue. The stem network blocks are represented by mild red. The deep network includes five, ten, and five blocks of variants, A, B, and C, respectively. In addition, a Reduction-A block (represented by pink) connects the Inception-ResNets A and B. In contrast, the Reduction-B block (represented by violet) connects Inception-ResNet-B and C. Pre-trained models are an efficient choice for classifying imbalanced medical data, as these models have optimized model parameters previously trained on large image datasets such as ImageNet. Further, re-training the model for the custom dataset optimizes their parameters. The modifications carried out to the base model are represented by colourless blocks in the Inception-ResNet-V2 block (left-end). The Modified Inception-ResNet-V2 is defined by replacing the topmost layer with an average pooling layer, followed by a flatten layer, dense layer with rectified linear unit (ReLU) activation, dropout layer, and output dense layer with SoftMax activation. Introduction of pooling layer reduces the size of the feature map, which is significant for capturing the most important features readily. Specifically, average pooling works by dividing its input feature map into (N) number of non-overlapping blocks. The pixel values pertaining to each block is calculated. These average values together form the new

Class description	Number of samples		
	Before splitting	Training data	Testing data
Malignant Melanoma	600	480	120
Benign Nevi	465	372	93

Table 1. Dataset details & Samples in each class.

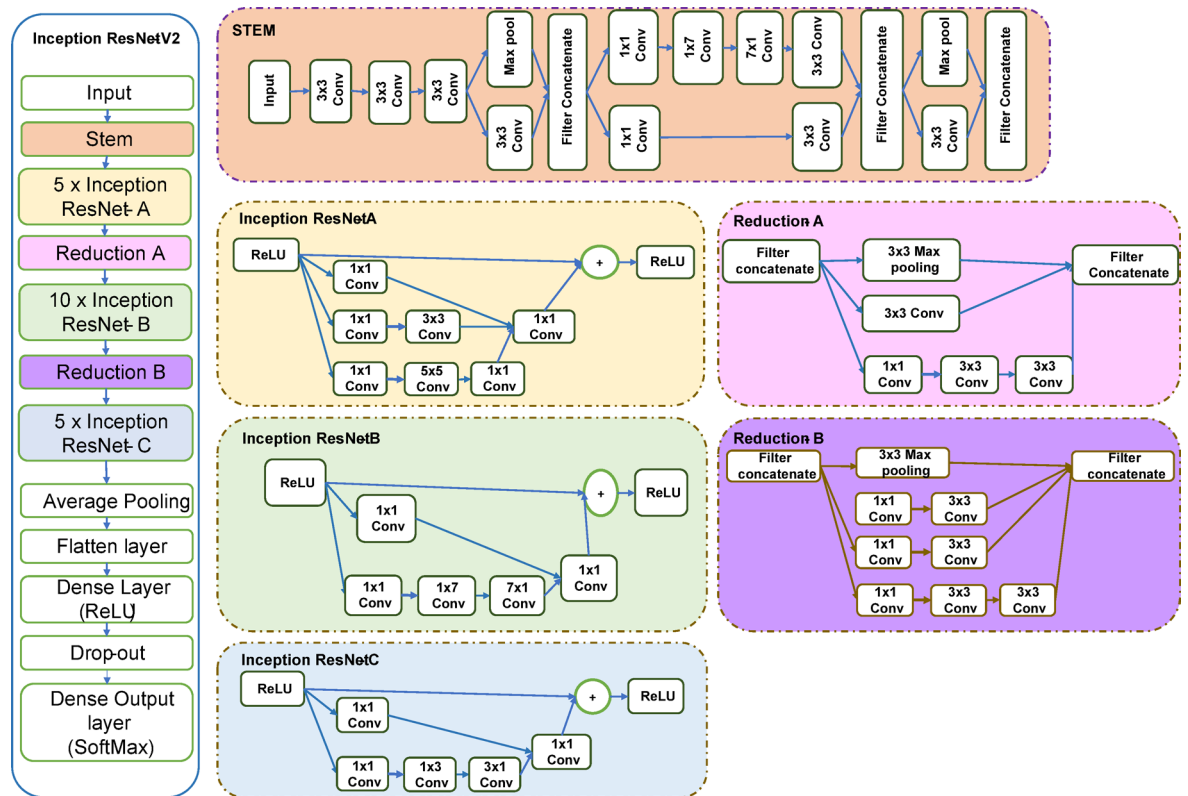


Fig. 5. Architecture of inception-ResNet-V2.

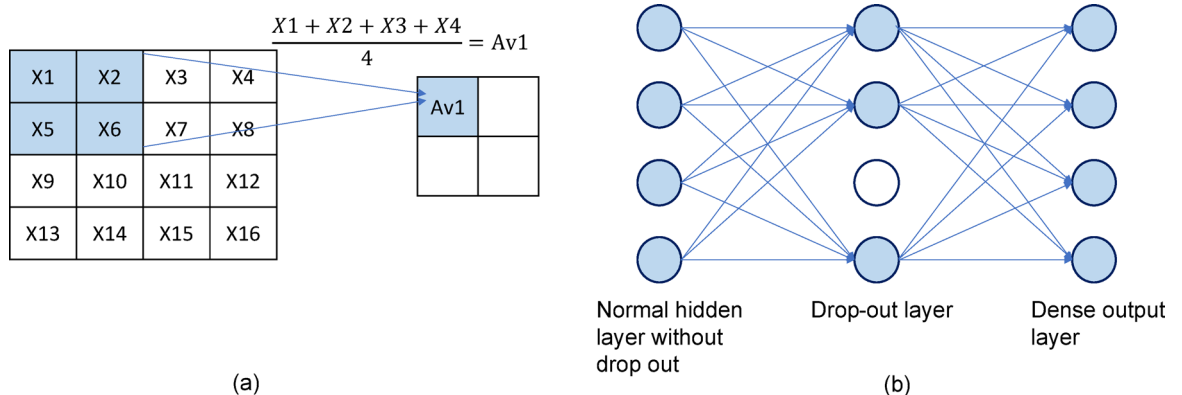


Fig. 6. Schematic representation layers in Modified Inception-ResNet-V2 model, (a) averaging pooling layer functions, (b) drop-out layer functions.

feature map. Thus, the feature map that results from average pooling will have only 'N' pixels in it. Diagrammatic representation of working of average pooling layer with a numerical example of average pooling function is given in Fig. 6a. There are two parameters that determine the size and value of the resulting feature map. One is the kernel size which depicts the number of pixels to be averaged and the stride which depicts the number of pixels to skip after each average calculation.

The pooling layer is followed by a flatten layer to convert the feature map to a single vector and then is fed to the dense layer. This dense layer and the output dense layer are connected through a drop-out layer in order to force a robust learning. The dropout layer randomly disables some of its neurons in order to force the network to learn robustly. The dropout layer randomly deactivates a fraction of its neurons in each iteration and the modifies network structure dynamically during each iteration in order to ensure a robust learning and model generalisation. A schematic of the drop out function is given in Fig. 6b. While SoftMax function transforms the raw scores obtained from the final layer of the network into class probabilities which helps in interpreting the

confidence interval of each class. These modifications are presented in the Inception-ResNet-V2 block in Fig. 5. These modifications are also shown in the overall block diagram in Fig. 4.

Model optimisation & Hyperparameter tuning

The proposed modified Inception-ResNet-V2 model is optimized using three optimizers, namely Adam, Nadam, and AdaMax, along with fivefold cross validation, to study the impact of optimization and tuning on the model performance. Validation carried out at the end of each training epoch to ensure model generalisation to unseen data. Training phase involves training models of different hyperparameters (here learning rate) to obtain multiple models each with their own weights and biases. Validation involves holding out a portion of training dataset and testing each of this model on the validation data. In this process, validation accuracy is compared to tune the model parameter (weight & biases) of each of the model (with their own specific hyperparameter) and the best performing model, which holds the most optimal model weights and hyperparameters is saved for evaluation phase. In simple programming terms this resembles a two level loop function where the hyperparameters are control variables in outer loop and model parameter namely weights and biases are control variables of inner loop. This validation phase introduces unseen data to the model, and checks model performance to unseen data. This process is repeated until there is no improvement in validation accuracy or till the maximum number of epochs, whichever is earlier. This study uses fivefold cross validation technique along with random search based hyperparameter tuning in order to obtain the best performing model. Further, as the input dataset is imbalanced, a stratified sampling is employed while splitting data. In the present study, the batch size and epochs are fixed and learning rate alone is tuned for efficacy. The model performance improvement with optimization and tuning, and the results in comparison with other pre-trained models and existing DL models, are discussed in the next section. The flowchart representation of the proposed algorithm used in the skin cancer classification model is presented in Fig. 7.

The flowchart depicts the training, validation and testing strategy adapted with the proposed methodology. A stratified sampling adapted while splitting training, testing and validation sets. Data augmentation is employed only to the training data samples and the testing set is hold-out for ensuring a robust prediction.

Results

The experiment was conducted on an Intel Core i7 (gen12) processor with an NVIDIA GPU. The training dataset was highly imbalanced; hence, minority classes were oversampled to arrive at a more balanced dataset to avoid bias during learning. Eighty% of the data were used for training, while 20% were reserved for evaluating

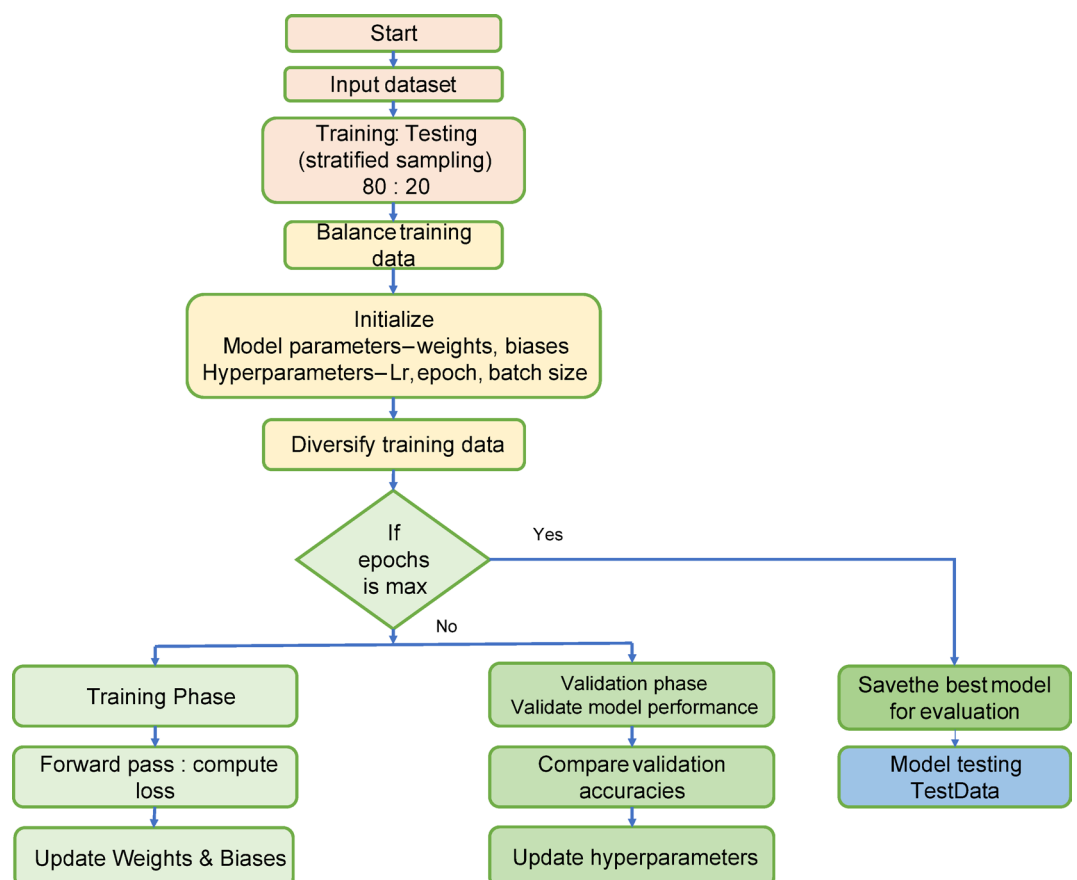


Fig. 7. Flowchart representation of the proposed methodology.

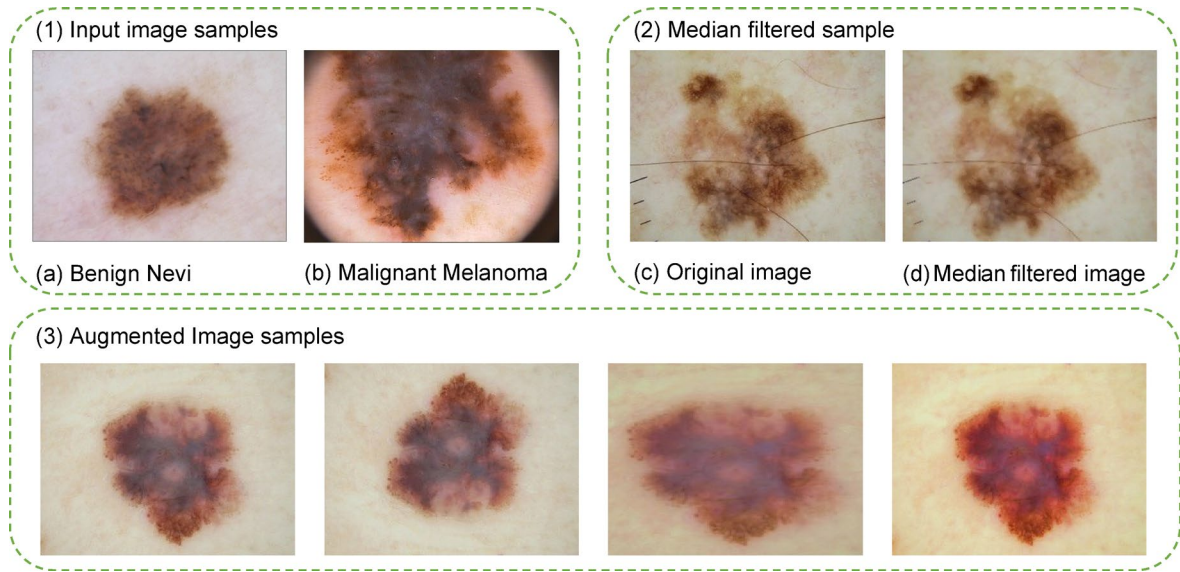


Fig. 8. Samples lesion images. (1) represents the sample input images, (a) benign nevi image, (b) malignant melanoma, (2) Median filtered sample, (c) input image (d) median filtered image, (3) sample images obtained after augmentation.

Performance parameter	Definition and formula
Accuracy	The ratio of total number of correct predictions to the total predictions. $\text{Accuracy} = \frac{TP+TN}{TP+FP+TN+FN}$
Sensitivity	The ratio of true positive predictions to actual positive cases (true positive rate) $\text{Sensitivity} = \frac{TP}{TP+FN}$
Specificity or recall	The ratio of true negative predictions to actual negative cases (true negative rate) $\text{Recall or Specificity} = \frac{TN}{TN+FP}$
Precision	The ratio of true positives to the predicted number of positives is calculated based on the positive expected outcomes. $\text{Precision} = \frac{TP}{TP+FP}$
F-measure	F-measure (harmonic mean of precision and recall), also called F1 score. $\text{F1 measure} = 2 * \frac{\text{precision} * \text{recall}}{\text{precision} + \text{recall}}$

Table 2. Performance metrics considered for evaluating the model.

model generalization. In addition, real-time data augmentation with 30° rotation, width shifting, height shifting, shear, zoom, and flip was performed to diversify the dataset. Figure 8a and b show samples of benign nevus and malignant melanoma lesions from the input dataset. The resemblance of these two images clearly depicts the difficulty associated with classifying them.

The results from pre-processing block are presented in Fig. 8(2) and 8(3). The images in Fig. 8c and d represents input sample and corresponding output sample obtained from a median filter with a window size of 5. It can be found from the filtered image in 8(d) that the hairlines appear blurred in comparison to the original image in Fig. 8c. This reveals the efficacy of median filters in removing noisy hairlines in spite of maintaining the required edge information. The augmented images are shown in block labelled (3) in Fig. 8. It could be found that augmentation resulted in producing images variants that mimic real-time image captures under varied lighting conditions and positions.

Performance metrics

The performance of the model is verified using various metrics. Each metric is calculated from four prediction measures: (i) True Positive (TP), number of correct positive predictions; (ii) True Negative (TN), number of correct negative predictions; (iii) False Positive (FP), number of positive predictions that are actually negatives; and (iv) False Negative (FN), number of negative predictions that are actually positives. The definitions of different performance metrics and their corresponding formulas are listed in Table 2.

These metrics provide data scientists and researchers with quantitative measurements of a model's performance against a specific dataset, enabling them to evaluate its advantages and disadvantages and make well-informed choices on model selection and optimization.

Experimental results

The Modified Inception-ResNet-V2 model was developed by replacing the topmost layer of Inception-ResNet-V2 and adding an average pooling, dense layer, dropout layer, and SoftMax layer. First, the classification efficiency

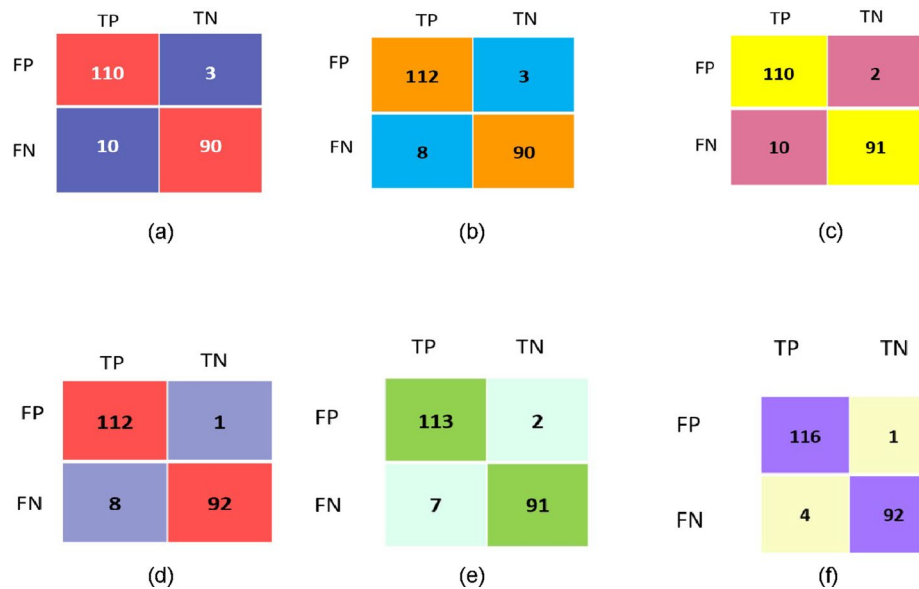


Fig. 9. Confusion matrix obtained with various model testing. (a) ResNET-50, (b) EfficientNet-B0 (c) Inception-v3 (d) Inception-ResNet-V2 + Adam (e) Inception-ResNet-V2 + Nadam (f) Inception-ResNet-V2 + AdaMax.

Model	Precision	Specificity	Sensitivity	Accuracy	F1 score
Resnet-50	97.35	96.77	91.67	93.9	94.42
Efficientnet-b0	97.39	96.77	93.33	94.87	95.32
Inception-v3	98.21	97.85	91.67	94.37	95.83
Inception-ResNet-V2 + Adam	99.12	98.92	93.33	95.77	96.14
Inception-ResNet-V2 + Nadam	98.33	97.85	95.16	96.24	96.72
Inception-ResNet-V2 + AdaMax	99.15	98.92	96.67	97.65	97.89

Table 3. Performance of various pre-trained models in classifying skin lesions.

of the modified Inception-ResNet-V2 model was compared with those of ResNet-50, EfficientNetB0, and Inception-V3. In addition, the effect of optimizers and hyperparameter tuning on the performance of the model was studied considering three optimizers, namely Adam, Nadam, and AdaMax and fivefold cross validation. Hyperparameter tuning employed a random search strategy and learningrate in the range of 0.001 and 0.0001 is searched. Validation accuracy is monitored and the best performing model is saved after validation. Other hyperparameters namely batch size and number of epochs are maintained constant at 32 and 40, and cross entropy is used as loss function. The best performing model after validation used a learning rate of 0.001. The proposed model got trained in 36 min when run on a Intel Core i7 (gen12) processor with an NVIDIA GPU, 16 GB RAM. The model is tested on distinct test data which is not used for training and validation. The confusion matrix obtained from model testing on hold-out test data are presented in Fig. 9a–f below.

Confusion matrices in Fig. 9a, b and c are obtained for ResNET-50, EfficientNet-B0 and Inception-v3 respectively. While Fig. 9d, e, f represent the confusion matrices obtained for the three variants of the modified Inception-ResNet-V2 model which uses Adam, Nadam and AdaMax optimisers respectively. It is inferred that from Fig. 9, that the false predictions obtained with the standalone models namely, ResNET-50, EfficientNet-B0, and Inception-v3 are more when compared to the false predictions made by hybrid Inception-ResNet-V2 variants. This reveals the robustness of the hybrid model to unseen data. These results are supported by the reduced sensitivity (from Table 3) with standalone models in comparison to hybrid counterparts. Further, model testing results from hybrid models exhibited a consistent value above 95% across the performance parameter like sensitivity, specificity and accuracy which clearly depicts model generalisation to unseen data.

The various performance metrics of the developed model compared with other pretrained models and different optimizations is presented in Table 3. A bar chart representation of model accuracy is presented in Fig. 10. Statistical evaluation of the results obtained is carried out to understand the reliability of the predictions. Clopper-Pearson confidence Interval (CP-CI) is calculated for the accuracy obtained with model testing. The CP-CI value obtained for ResNet-50, EfficientNet-B0, Inception-V3 are (89.7%, 96.5%), (91.42%, 97.12%), (90.49%, 97.02%) and those obtained with hybrid model (92.54%, 97.60%), (92.79%, 93.29%) and (94.4%, 99.1%) respectively for Adam, Nadam and AdaMax optimisers. This reveals the predictions made by hybrid Inception-ResNet-V2 variants are reliable and consistent in comparison to standalone models.

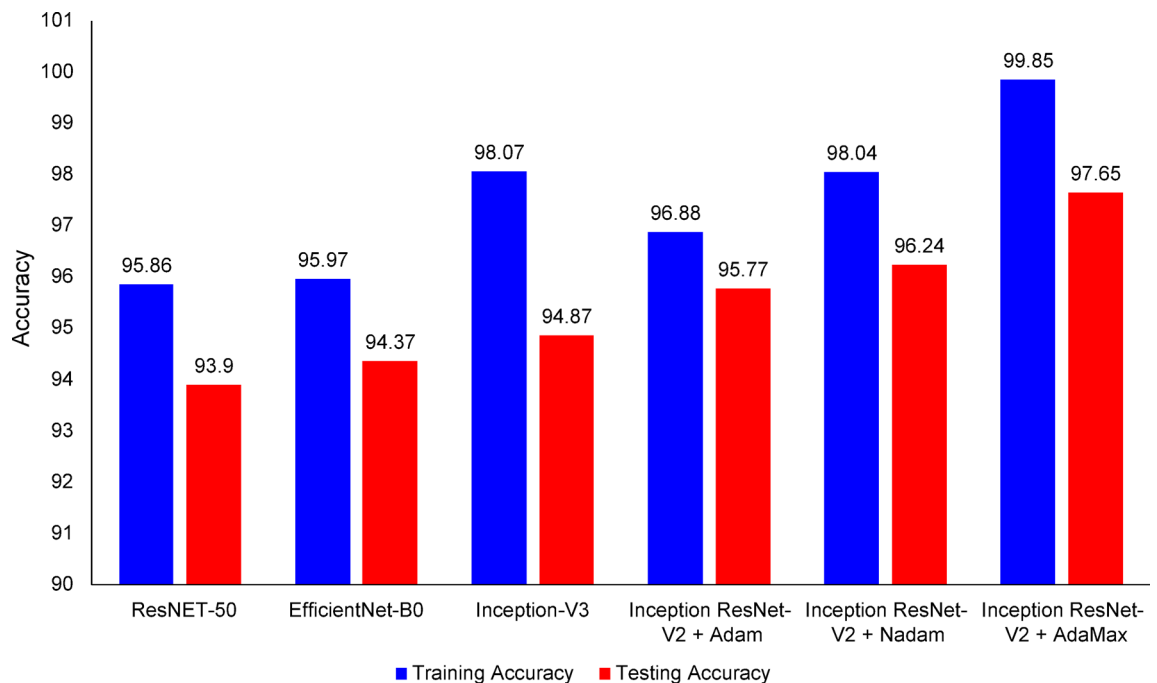


Fig. 10. Training and testing accuracy.

Comparison of area under the receiver operating characteristics curve

The AUC-ROC is a measure of how well a model can discriminate between classes. An ROC curve was obtained by plotting the TP rate against the FP rate for the different decision thresholds. The ROC plot has a 45°-reference line which indicates the model's differentiating power against random chance. The curve can be interpreted by studying the AUC range, the upper left corner of the curve, its curvature, and its closeness to the 45° reference line. An ROC curve with an AUC close to 0.5, will result in an ROC curve close to the reference line, indicating the model's inability to discriminate samples. An AUC in the range of 0.7–0.9 will appear well above the reference line and indicates that the model could discriminate between positive and negative samples well. Similarly, the upper left corner is close to 0.8, 0.9, and indicates good model performance. This curve creates a right-angle at the left corner. The ROC curves obtained while discriminating between malignant and benign classes using different models are presented in Fig. 11a–f.

The ROC curves obtained with the pre-trained models, ResNet-50, EfficientNet B0, Inception-V3, and optimized Inception-ResNet-V2 with Adam, Nadam, and AdaMax are presented in Fig. 11a–f, respectively. Every ROC curve is placed well above 0.5, which signifies that the pre-trained models are good at discriminating between positive and negative classes. This inference is supported by the ROCs curves above 45°. However, pre-trained models, when used directly, without tuning and optimization (i.e., with the same pre-trained weights), could achieve acceptable results ranging from 0.8 to 0.9, while classifying skin lesions. In the present experiment, ResNet-50, EfficientNetB0, and Inception-V3 achieved AUC of 0.824, 0.849, and 0.874, respectively; their corresponding ROC curves are shown in Fig. 11a, b, and c. By contrast, model tuning and optimization improved the performance of the modified Inception-ResNet-V2 model, as revealed by their AUC values. The ROC curves were almost right-angled and had AUC of 0.900, 0.950, and 0.975 for the tuned Inception-ResNet-V2 with Adam, Nadam, and AdaMax optimizers, respectively, as depicted in Fig. 11d, e, and f.

Model interpretability

The block diagram representations of the model architecture in Fig. 4 provides the basic interpretability to the black box model. It defines the various blocks and their corresponding functions on input samples. The measures taken to ensure unbiased predictions are revealed. In addition, the detailed model architecture in Fig. 5 enables understanding the in-depth architecture of the DL model and its efficacy in ensuring model generalisation. Further, the number of false predictions and true predictions made by the model on unseen test data are presented as confusion matrices in Fig. 9, which reveals the model's efficacy. Further, the CP-CI values present the reliability of model predictions. The first two components (Figs. 4 and 5) enable interpretability to model architecture while the confusion matrix and CI data enable interpretation of model predictions. However, there is scope for improving the interpretability of the results by including various Explainable AI methods like LIME (Local Interpretable Model-agnostic Explanations), SHAP or PDPs. These methods enable trust and accountability of the model predictions. These methods will be investigated in future enhancements.

Performance comparison with existing models

A comparison of the results of the proposed method with other state-of-the-art models in literature is presented in the following section. The models employed for skin cancer classification tasks are categorized into four

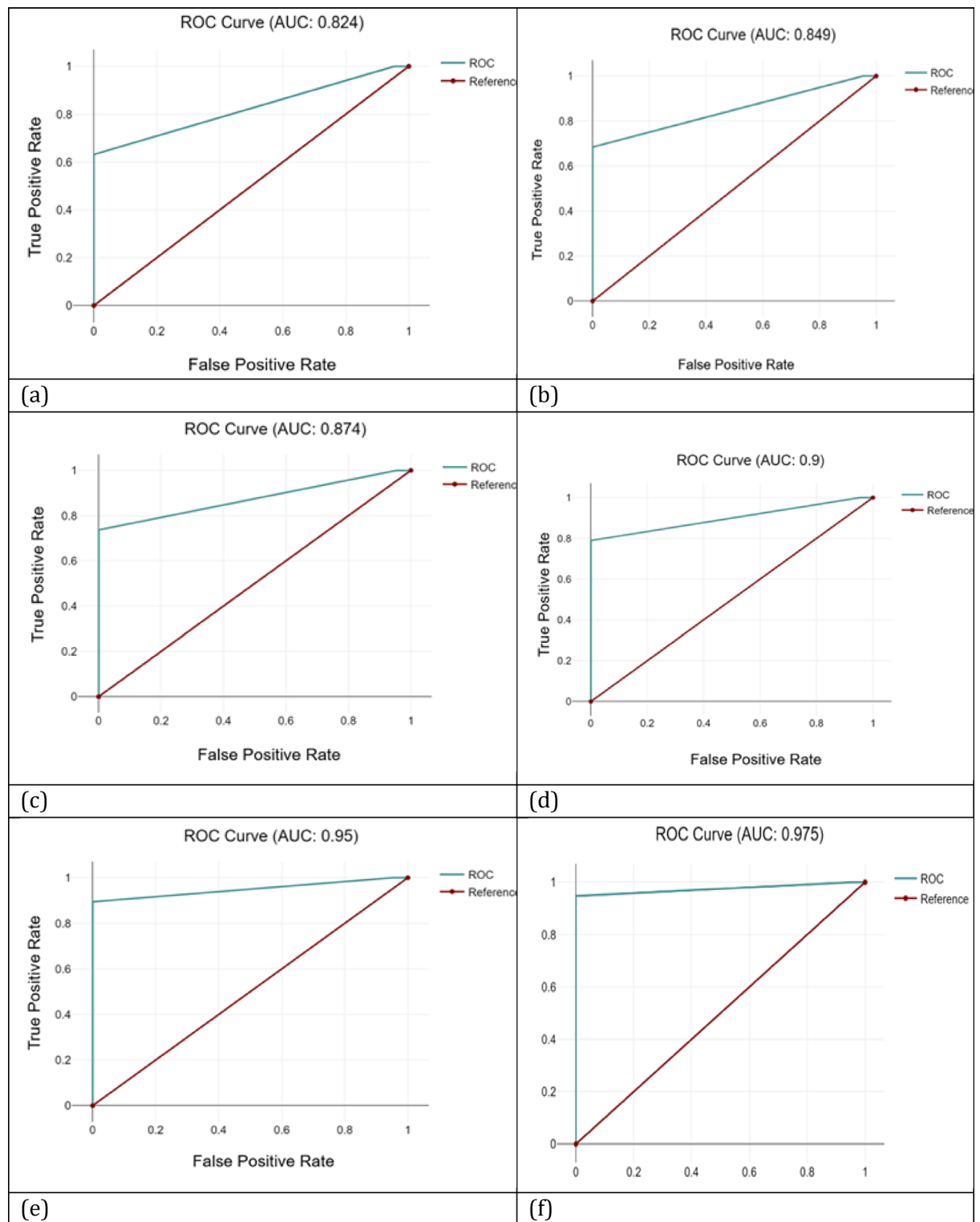


Fig. 11. ROC of various pre-trained models. (a) ResNET-50, (b) EfficientNet B0 (c) Inception-v3 (d) Inception-ResNet-V2 (e) Inception-ResNet-V2 + Nadam (f) Inception-ResNet-V2 + AdaMax.

major groups: (i) machine learning models that include separate feature extraction blocks, (ii) End-to-end CNN models, (iii) direct use of transfer-learning-based deep pretrained models, and (iv) tuned, optimized pretrained models. The results obtained from these four categories were compared to understand their effects on model performance. Table 4 lists the performance results obtained for the various models.

A comparison of the skin lesion classification results achieved with existing methods revealed that recently developed AI techniques can achieve an accuracy of more than 80% in classifying skin lesions. ML methods

References No	Model name / technique used	Accuracy	Precision	Recall / Sensitivity	Specificity	F1-score
Pre-processing, segmentation and ML classification						
14	Morphological filtering, Grabcut, GLCM, SVM	95.0	95.13	95.00	–	94.88
18	EW-FCM segmentation, Wide-ShuffleNet	82.56	–	82.56	97.51	–
17	Skin-Net (DermaNet)	94.65	72.56	70.78	96.78	71.33
End-to-end deep learning network models						
19	CNN	94	88	85	–	86
30	DKCNN	97.8	98.1	98.0	–	–
33	CNN + ANN	94.33	94.19	87.1	–	–
40	MSCD-Net	98.8	98.56	98.42	–	98.76
Transfer learning from pre-trained models						
23	Googlenet	81	77	74	84	–
25	Alexnet	95.91	92.32	88.47	93.00	–
Transfer learning + Tuned pre-trained models						
22	MobileNet V2 + LSTM	85.34	–	88.24	92.00	–
28	U-DenseNet121	95.8	–	93.5	97.1	–
28	U-MobileNetV1	96.0	–	93.2	97.2	–
31	DermaExpert	97	85	86	–	–
39	Vision Transformers	92.14	92.61	92.14	–	92.17
*	Proposed method	97.65	99.15	96.67	98.92	97.89

Table 4. Performance comparison of various models.

powered by an efficient segmentation stage and feature extraction stages produced an accuracy of approximately 94%, whereas DL methods produced a consistently higher accuracy. In particular, end-to-end deep learning networks tailored for skin lesion classification perform better than pretrained models. However, pretrained models can produce efficient results comparable to those of end-to-end models when tuned and optimized for skin lesion classification tasks. The proposed method achieved an efficiency similar to that of the existing models, although it was slightly lower than that of the tailored models.

Model significance, limitations and opportunities for extension

Clinical applicability and Integration to healthcare settings

- The proposed model enables enhanced skin cancer diagnosis because of its robustness architecture.
- It could support clinical decision making and helps clinicians in making informed decisions.
- In addition, the model when integrated with smart devices, facilitates remote access capabilities, acting as a virtual assistant.

Significance of the proposed model

- The most challenging case to distinguish in the ISIC 2019 dataset is benign nevi from malignant melanoma because of their similar appearances⁴¹. This study considers this challenging case and proposes a binary classification model for their precise classification and demonstrated to achieve excellent results.
- The proposed model included artifact removal and data augmentation techniques for data diversification. This enabled handling images obtained from practical settings which may include noises as well as are obtained at reduced perfection.
- Model optimisation and tuning are studied to understand their impacts on model performance as well as for obtaining robust and generalised model performance.

However, the model is not devoid of limitations and it holds numerous opportunities for improvement and extension.

Limitations

Though the present study proposed an efficient model, there are certain limitations which includes,

- The present methodology considered only a binary classification. It demands modification and tuning for multi-class classification which is required for more practical implementation and use.
- The model is trained on a single dataset namely ISIC 2019, and needs to be trained on other skin lesion datasets like ISIC 2020 or HAM10000 to ensure robustness.
- The model needs to be trained and tested on diverse datasets that carry different skin tones and images obtained from various imaging devices in order to ensure reliable integration to clinical predictions
- The model should be modified for handling 3D dermoscopic and volumetric data.

Scope for extension

- The standalone model will be modified for multi-class classification and trained on diverse datasets in order to facilitates a single model that is capable of providing complete skin cancer diagnosis.
- Explainable AI solutions will be included and user interface can be developed for improving usability of the model.
- Further, the model will be tested with images acquired from smartphone devices. Integration of model to smart device will be investigated for enabling remote diagnostic capabilities.

Conclusion

This study showed that the Inception-ResNet-V2 architecture combined with optimization methods, such as Adam, Nadam, and AdaMax, can effectively classify skin cancer stages. By utilizing the characteristics of deep learning and adjusting model parameters, it is possible to precisely differentiate between malignant and benign skin lesions. The models Inception-ResNet-V2 with Adam, Nadam, and AdaMax showed accuracies of 95.77%, 96.24%, and 97.65%, respectively. These results highlight the potential of the proposed model as a useful tool in dermatological applications. However, model robustness can be further improved by training and assessing the model with more diverse datasets. In addition, explainable AI can be introduced into the model using a suitable method. These requirements will be addressed in future studies.

Data availability

Data will be available by sending email to Corresponding author, satish Kumar with reasonable request.

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Author contributions

K. Umamathi—Conceptualization, editing, Supervision, writing original draft, Vanithamani R, Priya L, Harini K, satish Kumar—Conceptualization, editing, data curation, Mohsin Dadi, Saurav Dixit, Choon Kit Chan, Satish Kumar, Md Irfanul Haque Siddiqui—Supervision, writing original draft, funding acquisition.

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Declarations

Competing interests

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